

ITI Cmos Analog IC Design – Cadence Tools

Lab 03 Report — Cascode amplifier

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Part 1 — Sizing Chart

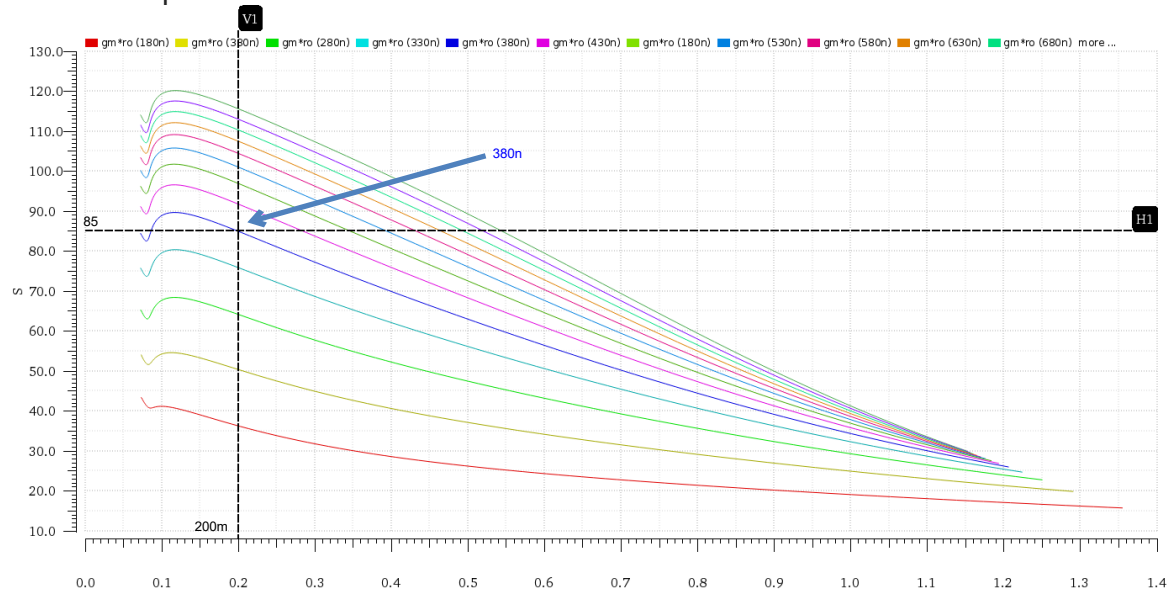
1.1 Design Specifications

Parameter	Value	Notes
IDQ (bias current)	70 μ A	Randomly generated
gm·ro (target intrinsic gain)	85	Randomly generated
V*	200 mV	Fixed
Supply (VDD)	1.8 V	0.18 μ m CMOS process

1.2 Determining L

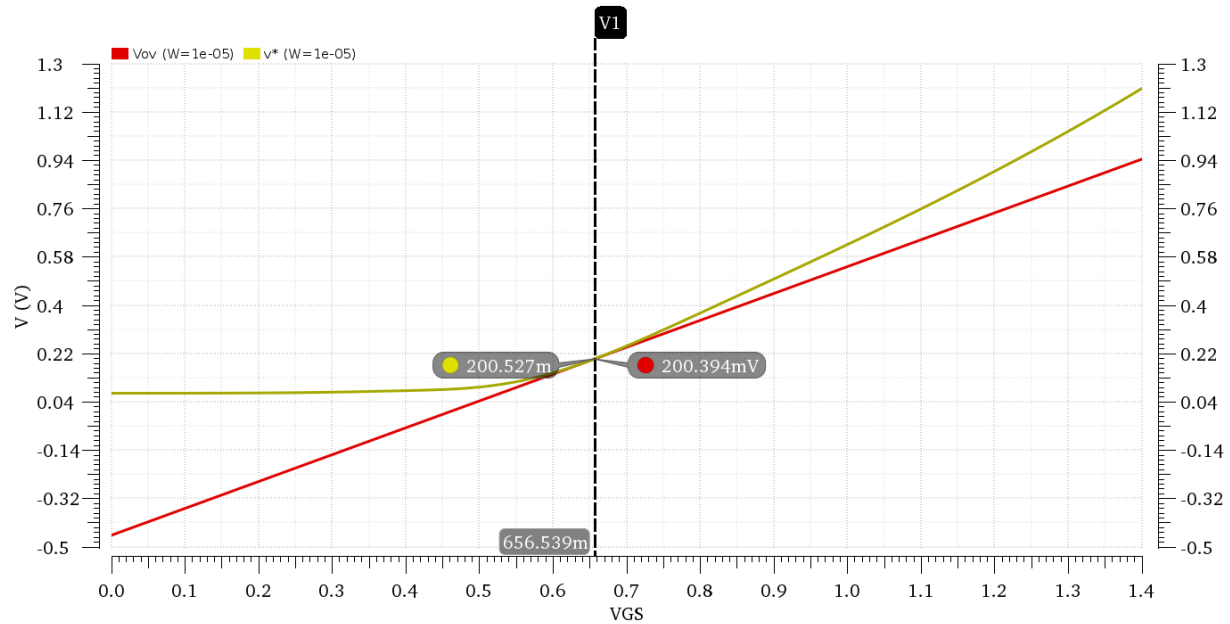
To determine the channel length L, weep was performed on both VGS and L. For each value of L, the intrinsic gain (gm·ro) was plotted VS V*. The required channel length L was then selected as the curve that satisfies gm·ro = 85 at the V* = 200 mV operating point.

➤ L-Sweep Results



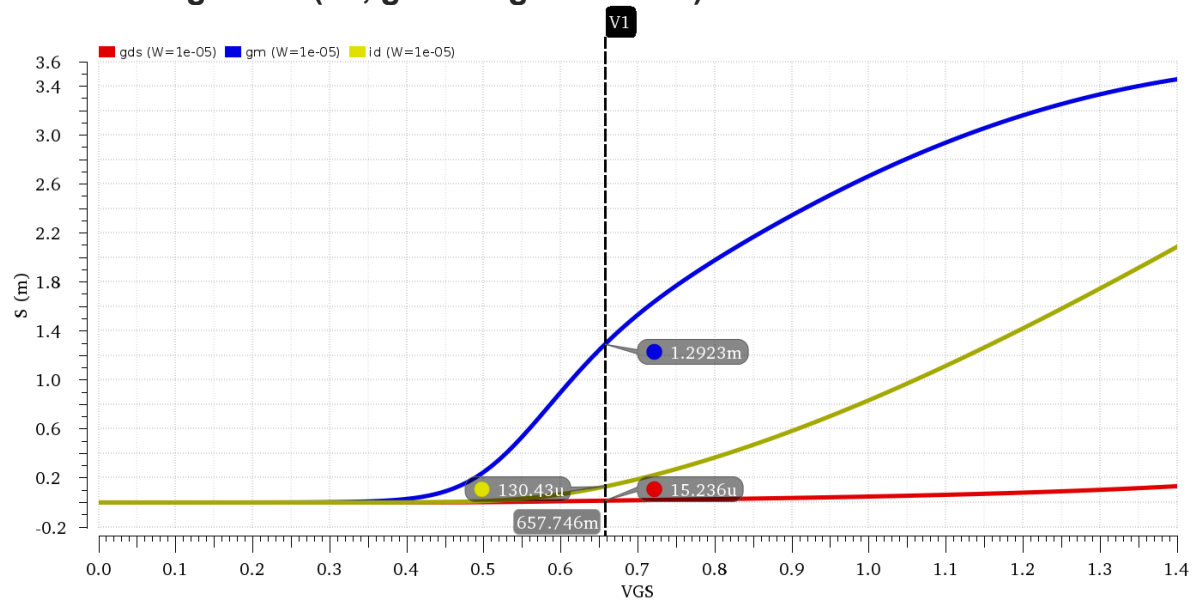
From chart L = 380 nm at V* = 200mV and gmro =85 .

2.3 V^* and V_{ov} vs. V_{GS}



From chart $V_{GS} = 656$ mV at $V^* = 200mV$

1.4 W- Sizing Chart (i_d , g_m and g_{ds} vs V_{GS})



From chart $i_d = 130$ uA, $g_m = 1.29$ mS and $g_{ds} = 15.26$ uS at $V_{GS} = 658$ mV

➤ Cross multiplication table

Since I_D is proportional to W at fixed L and V^* , the actual width is found by cross-multiplication:

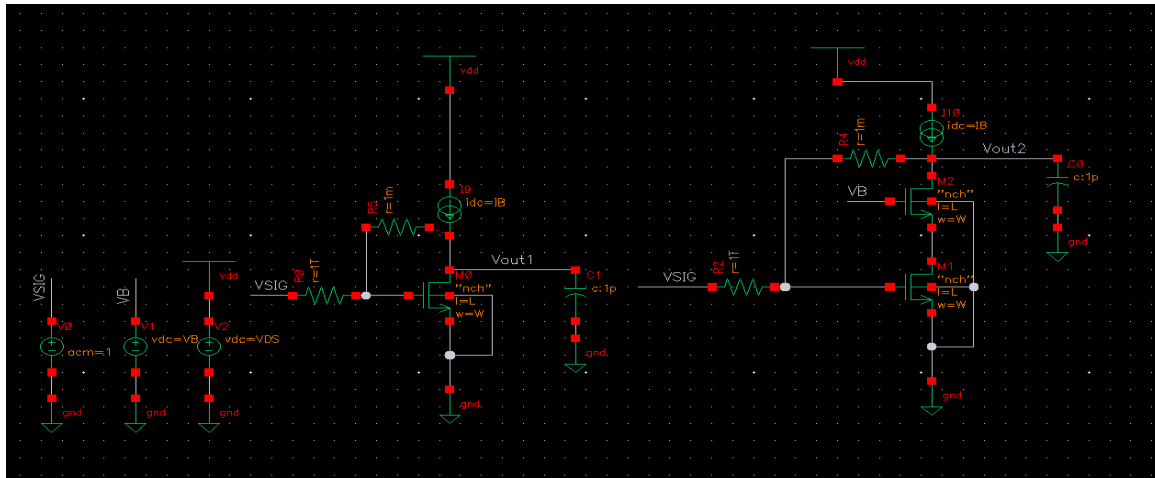
W	I_D	gm	gds
10 μm	$I_{DX} = 130 \text{ uA}$ (from chart)	$gm_X = 1.29 \text{ mS}$ (from chart)	$gds_X = 15.26 \text{ u}$ (from chart)
$W_Q = (10 \text{ u} * 70 \text{ u}) / 130 \text{ uA} = \underline{5.35 \text{ um}}$	$I_{DQ} = \underline{70 \text{ uA}}$ (spec)	$gm_Q = (1.29 \text{ m} * 70 \text{ u}) / 130 \text{ uA} = \underline{694.6 \text{ uS}}$	$gm_Q = (15.26 \text{ m} * 70 \text{ u}) / 130 \text{ uA} = \underline{8.2 \text{ uS}}$

$$G_m * r_o = gm / gds = 694.6 \text{ u} / 8.2 \text{ u} = \underline{84.7} \text{ (meet specs)}$$

Part 2 — Cascode for Gain

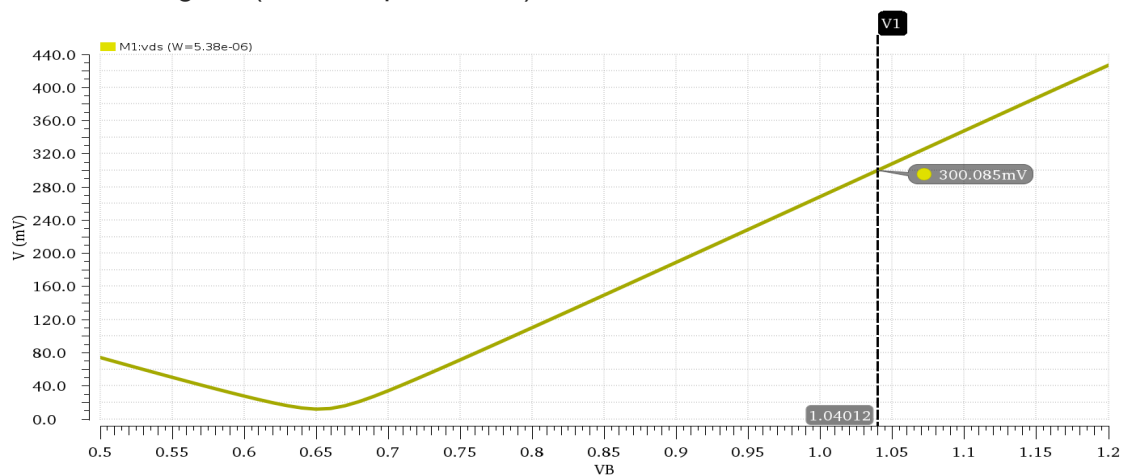
design specs

Parameter	Value	Notes
IDQ (bias current)	70 μ A	Randomly generated
gm·ro (target intrinsic gain)	85	Randomly generated
V*	200 mV	Fixed c
L	380 nm	From sizing chart (part 1)
W	5.35 μ m	From sizing chart (part 1)
Supply (VDD)	1.8 V	0.18 μ m CMOS process



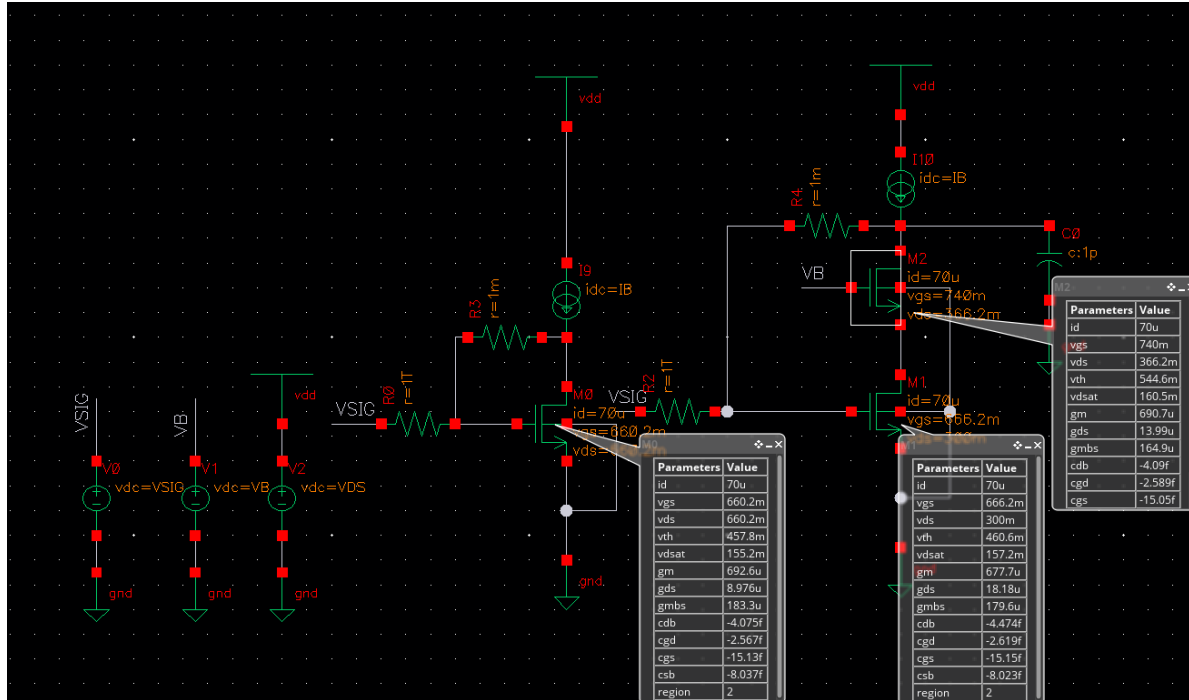
2.1 OP Analysis

➤ Choosing VB (VB sweep VS VGS)



From chart **VB = 1.04 V** At Vds = v* + 100mV = 300 mV

➤ DC Operating Point Analysis



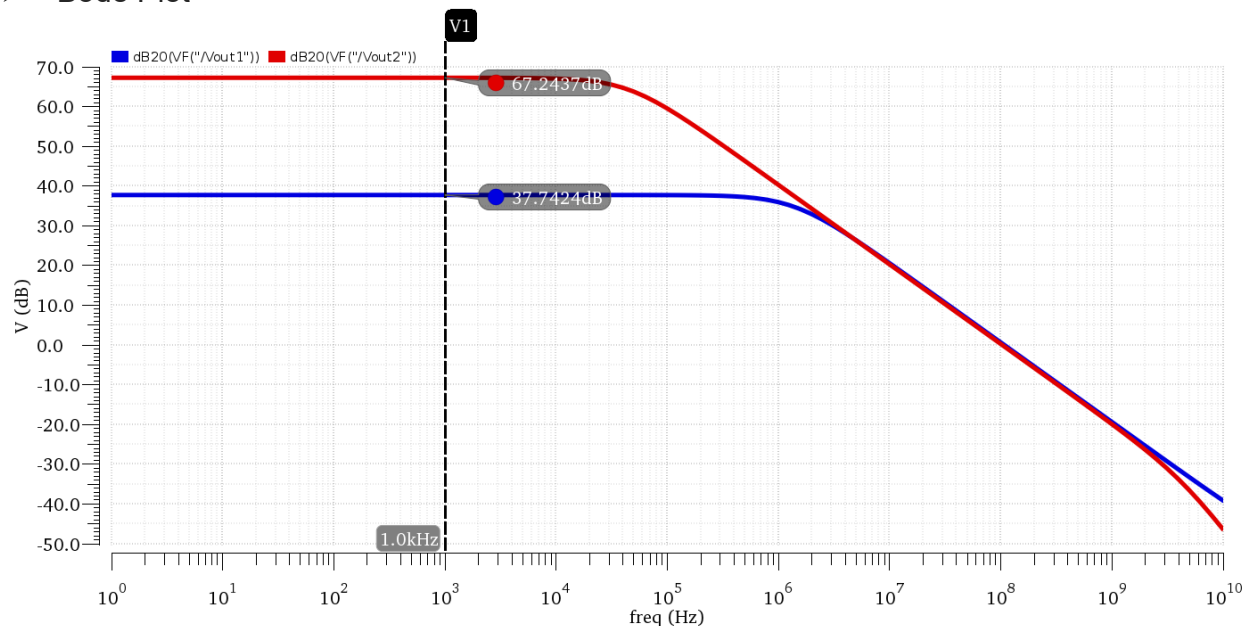
Parameter	M0 (cs)	M1	M2
ID	70 μ A	70 μ A	70 μ A
VGS	660.2 mV	666.2 mV	740 mV
VDS	660.2 mV	300 mV	366.2 mV
VTH	457.8 mV	460.6 mV	544.6 mV
VDSAT	155.2 mV	157.2 mV	160.5 μ S
GM	692.6 μ S	677.7 μ S	690.7 μ S
GDS	8.976 μ S	18.18 μ S	13.991 μ S
GMB	183.3 μ S	179.6 μ S	164.9 μ S
CDB	4.075 fF	4.474 fF	4.09 fF
CGD	2.567 fF	2.619 fF	2.589 fF
CGS	15.13 fF	15.15 fF	15.05 fF
CSB	8.037 fF	8.023 fF	7.158 fF
Region	2 (saturation)	2 (saturation)	2 (saturation)

- All transistors operate in saturation (region 2)

- M0 and M1 are very close (458–461 mV) in v_{th} , only just due to different V_{DS}/V_{GS} bias points , But M2's V_{TH} is higher (544.6 mV) because of the body effect ($V_{SB} \neq 0$), its source not grounded.
- For all transistor g_m **is much larger** that g_{ds} and that's expected and make g_m/g_{ds} is larger.
- For all transistor g_m **is larger** than g_{mb} ($g_{mb} = 0.2-0.3 * g_m$)
- For all transistor c_{gs} **is larger** than c_{gd} and that make sense due to all transistor operate in saturation region, which is gate-source area is much greater that gate-drain area.
- For all transistor c_{sb} **is larger** than c_{db} , and In saturation the source-body junction is more wider than the drain-body junction ,which increase junction capacitance.

2.2 AC Analysis and Result

➤ Bode Plot



labs:cascode:1	dB20(VF('/Vout1'))				
labs:cascode:1	ymax(dB20(VF('/Vout1')))	37.74			
labs:cascode:1	ymax(mag(VF('/Vout1')))	77.11			
labs:cascode:1	bandwidth(VF('/Vout1') 3 "lo...	1.417M			
labs:cascode:1	gainBwProd(VF('/Vout1'))	109.5M			
labs:cascode:1	dB20(VF('/Vout2'))				
labs:cascode:1	ymax(dB20(VF('/Vout2')))	67.25			
labs:cascode:1	ymax(mag(VF('/Vout2')))	2.303k			
labs:cascode:1	bandwidth(VF('/Vout2') 3 "lo...	45.44k			
labs:cascode:1	gainBwProd(VF('/Vout2'))	104.9M			

- The c_s gain in stimulation decrease from 85 to 77 because the V_{ds} of the transistor differ .66 V from the v_{ds} of sizing chart which is 0.9 V , so g_m and g_{ds} changed and affect on gain as I mentioned.

➤ Hand Analysis

Using the small-signal parameters extracted from the OP simulation

for CS

$$A_v = g_m \cdot r_o = 692.6 \mu / 8.97 \mu = 77.2 \text{ (37.8 dB)}$$

$$BW = \frac{1}{2 * \pi * C_l * \frac{1}{g_{ds}}} = \frac{1}{2 * \pi * 1 p * 111.4 k} = 1.43 \text{ MHz}$$

$$GBW = A_v * BW = 77.2 * 1.43 \text{ M} = 110.4 \text{ MHz}$$

for casode

$$R_{out} = g_{m2} * r_{o2} * r_{o1} = 690.7 \mu * 71.47 k * 55 k = 2.72 \text{ M}\Omega$$

$$A_v = g_{m1} * R_{out} = 677 \mu * 2.72 \text{ M} = 1843.08 \text{ (65.8 dB)}$$

$$BW = \frac{1}{2 * \pi * C_l * R_{out}} = \frac{1}{2 * \pi * 1 p * 2.72 \text{ M}} = 58.5 \text{ kHz}$$

$$GBW = A_v * BW = 1843.08 * 58.5 \text{ k} = 107.8 \text{ MHz}$$

➤ Simulation vs. Hand Analysis Comparison

Metric	CS		Cascode	
	Simulation	Hand Analysis	Simulation	Hand Analysis
DC Gain (dB)	77.11 (37.0 dB)	77.2 (37.8 dB)	2330 (67 dB)	1843.08 (65.8 dB)
Bandwidth (Hz)	1.4 MHz	1.43 MHz	45 kHz	58.5 kHz
GBW (Hz)	104.9 MHz	110.4 MHz	104.9 MHz	107.8 MHz
UGF (Hz)	110.6 MHz	110.4 Mhz	107.8 Mhz	107.8 MHz

- hand analysis and simulation agree for the cs amplifier within(5%) while the cascode shows larger error (23%) and that because to simplifying assumption in the Rout .
- The cascode increase dc gain with x30 and decrease bandwidth 31% and that conforming the gain bandwidth trade off.
- GBW and UGF almost remain constant for both circuit .

